Design Note | Pune Residential Apartment Ranking Tool

BRICKHUB INTELLIGENCE | Pre-Build Architecture & Methodological Blueprint

HYBRID DECISION PROTOTYPE — Verified Public Infrastructure + Calibrated Market Deed Records

1. Decision Context & User Persona

This intelligence tool supports an institutional Investment Committee at a mid-sized real estate private equity firm. The committee must decide which Pune micro-markets to prioritize for capital deployment. The target investor persona seeks a 5-to-7-year buy-and-hold residential apartment investment, demanding an optimal blend of medium-term capital appreciation and recurring gross rental income. The tool acts as a disciplined, repeatable first-stage screening engine.

2. Coverage Scope & Locality Selection Rationale

The scope covers exactly 12 Pune micro-markets: Baner, Wakad, Kharadi, Hadapsar, Hinjewadi, Balewadi, Bavdhan, Aundh, Wagholi, Pimple Saudagar, Kothrud, and Undri. This selection provides geographic diversification across the Western IT corridor (Hinjewadi, Wakad, Baner, Balewadi), Eastern commercial corridor (Kharadi, Hadapsar, Wagholi), mature prime central-west nodes (Kothrud, Aundh), and developing southern/western peripheries (Undri, Bavdhan). It captures both mature and emerging high-growth catchments.

3. Indicator Selection Rationale

Seven core metrics were chosen to mirror the investor mandate:
• YoY Transaction Price Growth (25%): Direct gauge of historical capital appreciation momentum.
• Demand-to-Supply Absorption Ratio (20%): Ratio of 12M units sold to units launched, identifying areas with structural undersupply.
• Gross Rental Yield (20%): Annual rental cash flow return on invested capital.
• Upcoming Infrastructure Catalysts (15%): Forward-looking rerate potential driven by verified transit infrastructure (Metro Line 3, Ring Road).
• Months of Unsold Inventory (10%): Lower overhang protects against developer price-cutting.
• Transaction Liquidity (5%): Secondary market sales velocity ensuring an easy exit.
• Risk & Valuation Affordability (5%): Downside protection penalizing developer litigation and excessive price premiums.

4. Combination Method Justification

We employ a Multi-Criteria Decision Analysis (MCDA) framework with indicator-wise Min-Max normalization (0–100 scale) and linear weighted sum scoring. For small sample sizes (N = 12), black-box machine learning or unsupervised clustering models risk severe overfitting and fail to reflect subjective committee trade-offs. A weighted linear model provides complete auditability, transparent sensitivity testing, and direct scenario alignment.

5. Anticipated Data Issues & Planned Handling

• Missing Data: Handled deliberately via three selectable strategies: Median imputation (neutral baseline), Conservative 10th/90th percentile imputation (stress-test penalty), and Complete-case analysis.
• Data Calibration: Real-world public benchmarks (MahaRERA quarterly filings, IGR rate trends, PMRDA updates) anchor macro indicators; granular transaction rows are calibrated to reflect these observed market medians without violating API scraping constraints.